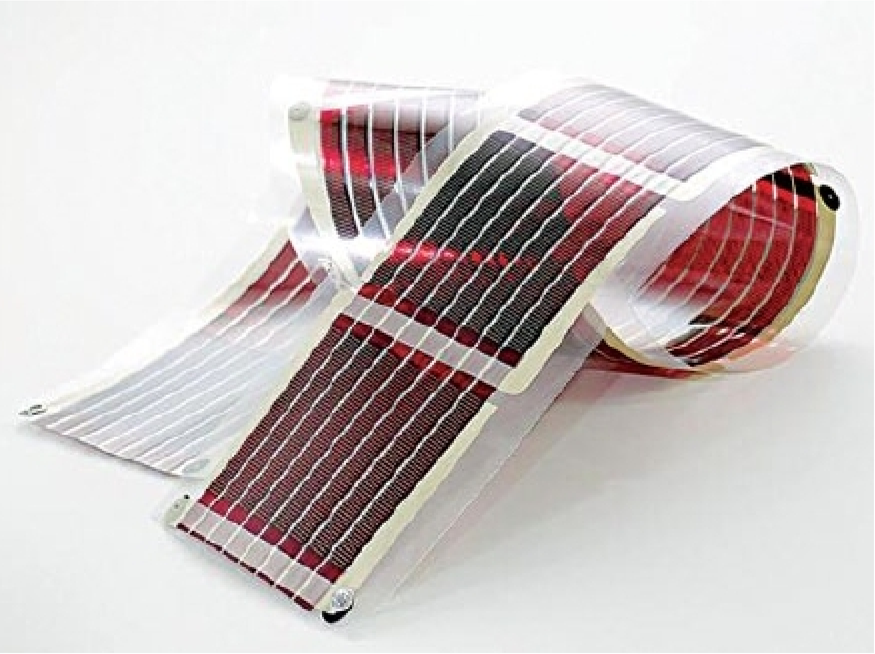

Fig. 2: Typical structure of a flexible organic solar cell

Fig. 3: A view of transparent organic solar cells installed on a building

(nanomaterial carbon) and P3HT polythiophene, which is basically a polymer.

A common characteristic of small molecules (fullerene) and polymers (P3HT) (Fig. 5), used as the light-absorbing material in photovoltaic, is that both have a large conjugated system. A conjugated system is formed where carbon atoms covalently bond with alternating single and double bonds. These hydrocarbons' electrons pz orbital delocalise and form a delocalised bonding π orbital with a π* ant bonding orbital. The delocalised π orbital is the highest occupied molecular orbital (HOMO), and π* orbital is the lowest unoccupied molecular orbital (LUMO).

In organic semiconductor physics, HOMO takes the role of the valence band, while LUMO serves as the conduction band. Energy separation between HOMO and LUMO energy levels is considered as the band gap of organic electronic materials, and is typically in the range of leV to 2.3eV.

All light with energy greater than the band gap of the material can be absorbed, though there is a trade-off for reducing the band gap, as photons absorbed with energies higher than the band gap will thermally give off their excess energy, resulting in lower voltages and power conversion efficiencies. When these materials absorb a photon, an excited state is created and confined to a molecule or a region of a polymer chain.

Excited state can be regarded as an exciton, or an electron-hole pair bound together by electrostatic interactions. In photovoltaic cells, excitons are broken up into free electron-hole pairs by effective fields. Effective fields are set up by creating a heterojunction between two dissimilar materials. In organic photovoltaic, effective fields break up excitons by causing the electron to fall from the conduction band of the absorber to the conduction band of the acceptor molecule.

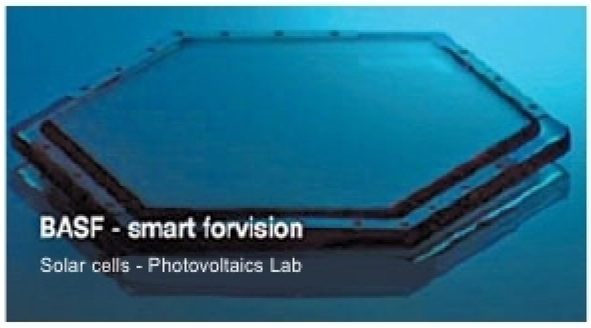

Fig. 4: BASF's organic photovoltaic structure

It is necessary that the acceptor material has a conduction band edge that is lower than that of the absorber material.

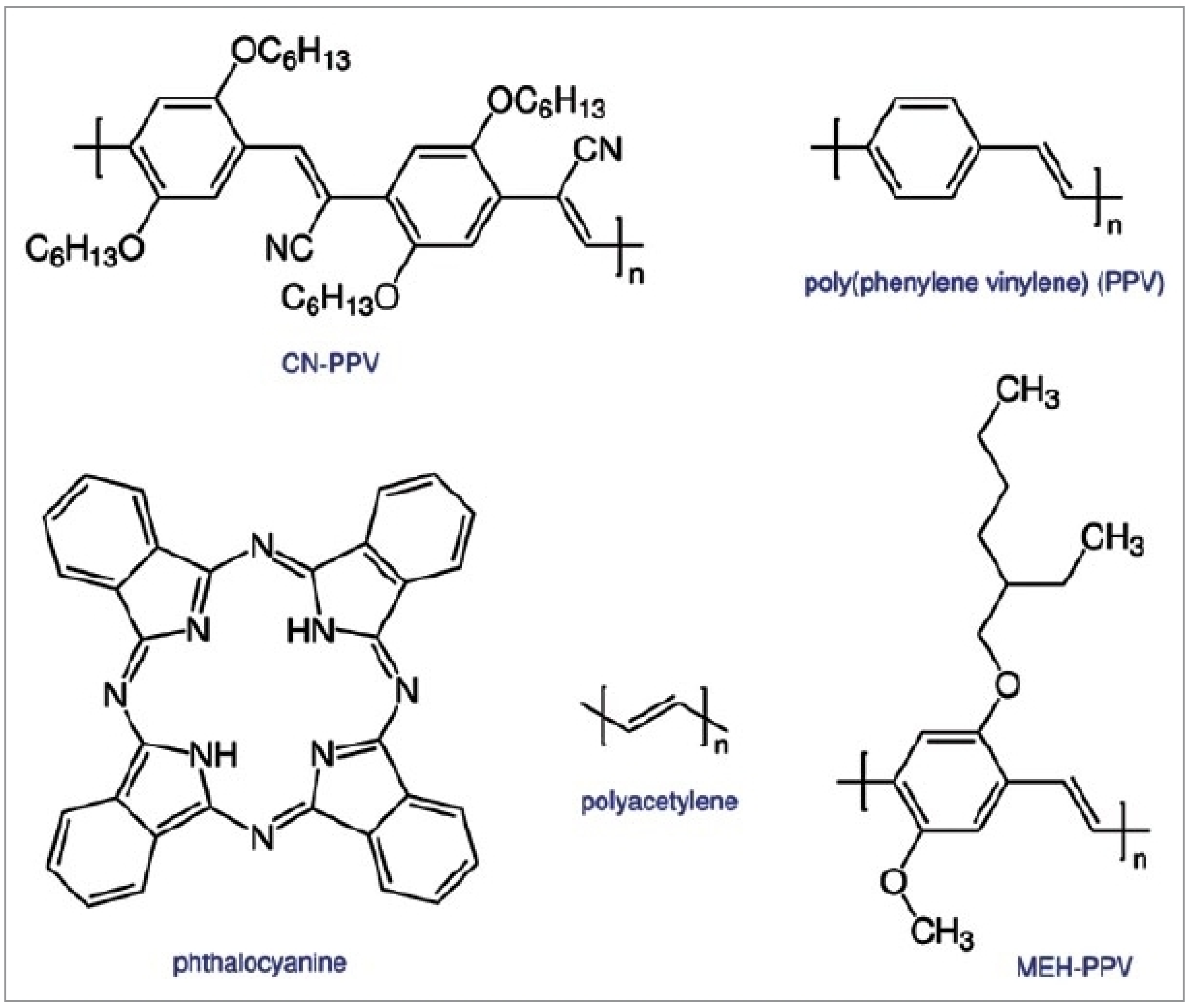

Fig. 5: Organic photovoltaic structure